Lumped-Parameter Modeling and Calibration of Hexapole Electromagnetic Actuators for Magnetic Flux Generation

Current Base Model

$$b_{ij} = \underbrace{\left(\tfrac{6}{5}k_{11}\right)\left(\frac{N_c}{4\pi\widehat{\ell}^2 R_a}\right)}_{{}^B\widehat{g}_I} S_i \overline{K}_I I = S_i\left(\underbrace{{}^B\widehat{g}_I \overline{K}_I I}_{G}\right)$$

Voltage Base Model

$$b_{ij} = S_i \cdot D \cdot V = S_i \cdot \left[\underbrace{\left(\tfrac{6d_{11}}{5}\right)}_{{}^B\widehat{g}_V}\overline{D}V\right] = S_i\left(\underbrace{{}^B\widehat{g}_V \overline{D}V}_{G}\right)$$

Symbol	Unit
R_a	$\frac{A}{Wb}$
I	A
${}^B\widehat{g}_I$	$\frac{mT}{A}$
${}^B\widehat{g}_V$	$\frac{mT}{mV}$
V	mV
$\widehat{\ell}$	μm
b_{ij}	mT
$\begin{cases} k_m = \frac{\mu_0}{4\pi} = 10^{-7} \\ \mu_0 = 4\pi \cdot 10^{-7} = 0.0013 \cdot 10^{-3} \end{cases}$	$\frac{N}{A^2}$